Verification of the Stefan-Boltzmann law as well as the Lambert law

Helke Parussel and Maria Scholl

A Leslie cube and a cavity radiator were used separately to test the T^4 dependence of the Stefan-Boltzmann law. The infrared transmittance of four different filters was investigated. Among them were filters made of window glass (SiO_2) with a transmittance of $(78.07 \pm 0.43)\%$, NaCl with $(46.26 \pm 0.24)\%$ and silicium with $(3.613 \pm 0.020)\%$. The angular dependence of Lambert's law was checked for correctness with a cavity radiator and declared to be approximately valid with the method used.

Execution of experiment: 6th of april 2022
first submission: 13th of april 2022

1 Introduction

Every substance, every body emits electromagnetic waves in the form of thermal radiation. In nature, this radiation is mostly in the wavelength range of infrared radiation. However, for some objects, such as the sun, the radiation maximum is in the ultraviolet range. The ability of materials to absorb and emit certain wavelengths is very important in parts of science. For example, this knowledge is used to insulate houses or to determine the location of a habitable planet by looking at the radiation emitted by the sun. In the following, the absorption behavior of material and surface texture will be investigated. The Stefan-Boltzmann- Law, which describes the emitted spectrum of a black body, is examined. The black-body radiation approached here is the radiation of a so-called black-body. This has the most effective thermal radiation emission and absorption in all wavelengths and temperatures and is used in theory in Planck's radiation law.

2 Theory

In the following sections, the emission behavior of different surfaces and that of a cavity radiator will be investigated. The optimum radiator is called a black-body radiator, which is capable of absorbing the entire spectrum of light and emitting it at the same time. The spectrum follows the intensity distribution of Plank's law of radiation:

$$\rho\left(\nu, T\right) d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} d\nu \tag{1}$$

Other bodies are black only in a certain frequency range, which means that only certain regions of the spectrum can be completely absorbed and emitted. The emitted power P of a black body is given via

$$P = \sigma A T^4 \tag{2}$$

and $\sigma = 5.67 * 10^{-8} \frac{W}{m^2 K^4}$. The intensity maximum of the radiance is given by Wien's displacement law using

$$\lambda_{max} = \frac{2897,8 \mu m\, K}{T} \tag{3}$$

A black body has a direction-independent as well as a constant radiation density on a surface. Because of this, the measured radiance decreases with decreasing angle according to Lambert's law:

$$J = J_0 \, cos\left(\varphi\right) \tag{4}$$

3 Observation of the radiation behavior of different bodies and surfaces

3.1 Investigation of the emission behavior of different surfaces with the Leslie cube

This part of the experiment deals with the emission properties of different surfaces of the Leslie cube. In total, there are four sides: black, white, dull and reflective, where reflective and dull are both made of the same material, namely nickel. The black and white sides are made of the material of the Leslie cube. A colored varnish was applied on their surfaces. The water, which is placed in the cube for heating, is heated, starting at a temperature slightly lower than room temperature. This temperature is monitored and controlled by the thermostat throughout the experiment. The thermal radiation from individual sides of the cube is then measured at selected temperatures up to 85°C using the thermopile and converted into a voltage difference. The connected digital multimeter (Agilent 34405A 1/2 digit) is used to display this thermo-electric voltage difference. The temperature of the cube and the corresponding voltage are noted. This is repeated for all four sides of the cube.

The temperature and thermoelectric voltage error are calculated according to the operating instructions of the LAUDA R 104 thermostat manual and the Aiglent 34405A voltmeter.

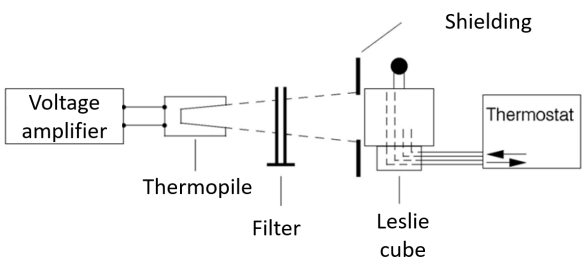

Figure 1: Set-up of the experiment to measure the emitting radiation of four different surfaces of the Leslie cube. The surfaces to be measured are black, white, mirror-finish and dull. Both of the latter are made of of nickel. In addition, this structure is also used for the investigation of the infrared permeability of different materials. The permeability was determined for window glass, NaCl, infrared glass and glass. Taken and edited from [3].

3.1.1 Exmperimental verification of the Stefan-Boltzmann law

Agilent 34405A 5 1/2 digit multimeter, KIPPZONEN CA thermopile and LAUDA Ecoline Staredition RE104 refrigeration thermostat were used in this part of experiment.
Figure 12 plots the measured thermo-electric voltage versus the actual temperature of the Leslie cube for the four sides. It has been considered that the temperature range extends from 15°C to 85°C in this part of the experiment. The dull and reflective sides are clearly below the black and white curves (see Fig. 12), which are very similar, almost identical. The dull and reflective sides are clearly below the other two. In the following, it is to find out whether all four sides follow this law.

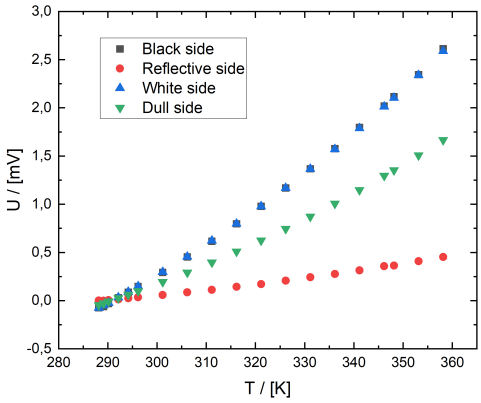

Figure 2: Measured thermo-electric voltages plotted against the actual temperature of the Leslie cube for the four different sides. Errors not plotted are too small for the chosen representation.

The measured thermo-electric voltage U_{Th} is described by the following formula:

$$\delta U_{Th} = k(T^b - T_0^b) \tag{5}$$

k represents a constant of proportionality that depends on the thermopile used and the environment b the exponent which the voltage is proportional to. As it is assumed that the voltage is proportional to the emitted radiation and therefore the exponent is expected to be $b = 4$. $T_0 = (291,132 \pm 0,066)$K was used as the reference temperature. It was determined from the data in Fig. 3 using the zero crossing. It is determined over the black side of the Leslie cube, since this side is assumed to be as close to a blackbody radiator as no other side of the cube. A value of $k = (4,4 \pm 1,1)10^{-10}$ V/K^4 is ascertained. The values for the exponent for all four sides are determined by the help of Fig. 3 and Fig. 4.

surface	α
black	(3.924 ± 0.040)
white	(3.92350 ± 0.00014)
reflecting	(3.63496 ± 0.00099)
dull	$(3,85001 \pm 0,00020)$

Table 1: Exponent b used for the verification of the Stefan- Boltzmann law

The eponent of the black side does not agrees with the expected power of $b = 4$ according to the Stefan Boltzmann law including error interval. However, the percentage deviation amounts to only 1.9%, which suggests that the black side is very close to a black body in practice, but is only an approximation. The white side with a similar exponent, as the black one, does not agree within the error with the theory as well. A confounding source of error is the T_0- drift of the reference temperature. Due to the lack of a cooling bath on the far side of the thermopile, the temperature of this becomes higher and higher in the course of the experiment, as the background radiation is influenced by the radiation of the Leslie cube. Subsequently the thermoelectric voltage becomes smaller. Hence, the evaluated exponent must be smaller than the theoretically expected exponent without this systematic error. Since the white side as well as the black one deviates only by 1.9% from the theoretical value, it can be concluded that this side also follows the Stefan- Boltzmann- Law as a good approximation.

xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx Both sides allow the thermal radiation to be transmitted similarly well, since they are made of the same material, namely the Leslie cube. However, this finding is unintuitive, since it was expected that the black side should have a higher absorption than the white one. In the visible spectral range, the white side is capable of reflecting almost all radiation, whereas the black side absorbs it. However, the emissive and absorptive power in the visible range does not translate to that in the infrared range. The potencies of the matte and reflective sides are far below those of the white and black. This can be attributed to the material they are made of: Nickel. If an electromagnetic wave hits one of the two sides made of nickel, a node is formed there in the electric field of the electromagnetic wave. A charge shift of electrons is induced in the metal, which counteracts the electric field. As a result, the infrared radiation is reflected and less is transmitted. The thermopile measures a lower ther-

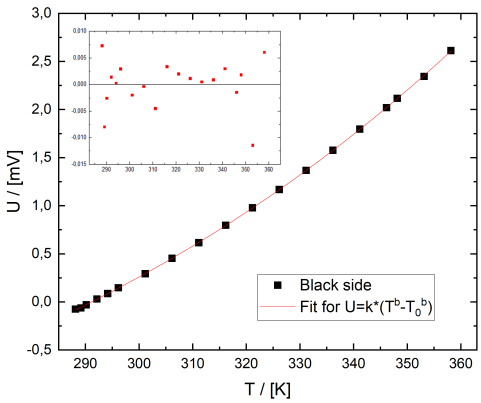

Figure 3: The black side of the Leslie cube is plotted over T. To check the Stefan Boltzmann law, the exponent is determined by the best fit of the data.

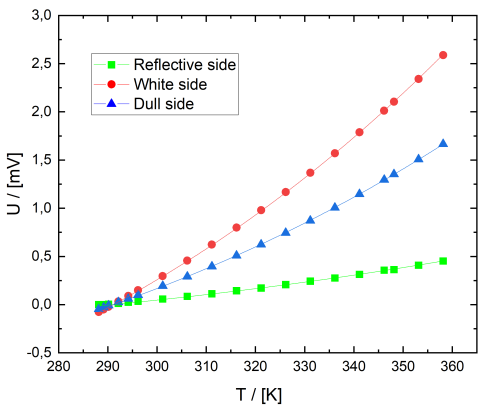

Figure 4: The sides white, dull and reflective are plotted over T. The Stefan- Boltzmann-law is checked with the help of the best fits and their determined exponent.

moelectric voltage than for the black and white sides. The fact that the matte side transmits more thermal radiation than the reflective side is due to its surface properties.

3.1.2 Ultra-red transmittance of different materials

The black side of the cube is used to examine the infra red transmittance of different substances. Here three different filters were used to represent those substances. At a temperature of (75.00 ± 0.40)°C, the thermoelectric voltage is first noted without a filter to obtain the maximum value of the radiation without a filter. Then the ambient radiation is measured with a nickel-plated filter that is inserted in front of the thermopile and does not transmit any radiation. Because this filter does not allow any radiation from the Leslie cube to pass through, it is possible to measure the thermoelectric voltage of the surrounding, possibly also reflected radiation. This measurement is carried out for the three remaining filters window glass (3 mm thick), a silicon plate (0.5 mm thick) and a plate of monocrystalline NaCl.

material	percent permeability
NaCl	(78.538 ± 0.038)
silicium	(35.991 ± 0.071)
window- glas SiO_2	(2.0755 ± 0.0012)

Table 2: Transmittance of infrared radiation of various materials at a temperature of (75.00 ±0.40)$°C$

The transmittance of these materials is determined as follows and plotted in Table 2:

$$\tau = \frac{U_F - U_0}{U_{0F} - U_0} \tag{6}$$

The device error of the voltage meter (Agilent 34405A 5 1/2 digit) in the 100mV measuring range was used to subsequently calculate the final error according to Gauss.
It is striking that window glass, which transmits the most light in the visible range, absorbs the most in the infrared. This is due to the non-crystalline, amorphous composition of glass. The molecules are excited to vibrate by the infrared radiation. The energy emitted from the black side of the cube can be calculated with the formula

$$E = \frac{hc}{\lambda} \tag{7}$$

T [K]	$\lambda_M [\mu m]$
291.132 ± 0.066	9.9536 ± 0.0023
348.15 ± 0.40	8.3234 ± 0.0096

Table 3: Wavelength λ_M of the emitted intensity maximum of the room temperature and that of the Leslie cube

and the value of the wavelength from Table 3 to calculate $E = (0.1490 \pm 0.0016)$eV. This energy is not sufficient to overcome the band gap of silicon (1.08eV) and NaCl (8.5eV) [4] [5]. The energy is too low and cannot be absorbed by the bound electrons. However, this is not the only reason for the high permeability of the two filters: The structural composition of the materials provides an insight into the infrared permeability. Silicon, like NaCl, has a crystalline atomic structure with strong interatomic bonds. In contrast to amorphous window glass, these two filters have a long-range order, i.e. there is a regular, periodic atomic order with constant bond angles. This structure is therefore more motionless in its form than the near order. This means that absorbed energies can be converted into atomic movements in the near order, but less so in the far order due to the fixed structure. The penetrating radiation is therefore transmitted with a higher proportion. Why electromagnetic waves transmit poorly in silicon, a metal, has already been explained in 3.1.1. This is not the case with sodium chloride, where the electron mobility is obviously lower than with a metal. Thus, its plasma frequency is much lower than that of the silicon plate. If the frequency of an electromagnetic wave is higher than the plasma frequency, no node forms on the plate because the electrons cannot oscillate fast enough, and the waves can transmit.
Finally, the corresponding wavelength of the intensity maximum of the room temperature and the temperature of the Leslie cube of this part of the experiment is determined according to Wien's displacement law using formula (3) and entered in Table 3.

3.2 Black-body radiation

A cavity radiator is used to investigate black-body radiation. It has almost perfectly absorbing walls, which are as impermeable to radiation as possible, and a small opening.
The temperature of the cavity radiator is controlled by a regulating transformer and measured by a thermopile (see Fig. 5). Since the temperature cannot be adjusted exactly, an error of $XYXYXYXYXY \pm 1$K is applied because the thermopile requires a certain adjustment time. The reading error of the angle is estimated at $XYXYXYXYX \pm 1°$.

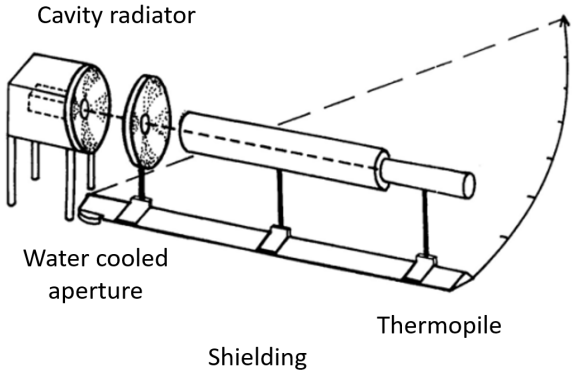

Figure 5: Set-up of the experiment to measure the temperature and angular dependence of an approximated black body using a cavity radiator. Taken from [3]

3.2.1 Stefan- Boltzmann law for cavity radiation

In this part of the experiment, the thermo-electric voltage difference U_{Th} was measured at an angle of $\varphi = 0°$. The cavity radiator was charged with a variable transformer. Voltages were measured between a temperature of 293K to 663K. The measured voltage was plotted against the temperature T in Fig. 6.

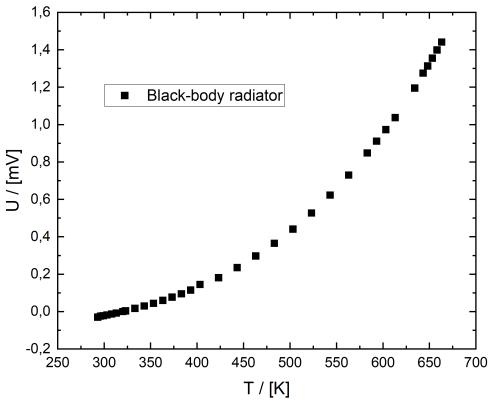

Figure 6: Measured thermoelectric voltage plotted against the temperature of the cavity radiator in a range from XK to XK. Errors which are not plotted are too small for the chosen representation

In order to verify the Stefan- Boltzmann law, a fit was adjusted to the graph in Fig. 7. The same reference temperature $T_0 = (291.132 \pm 0.066)\,K$ was used like in 3.1.
To compare if this method, used here and in 3.1, is reliable, the data was also evaluated differently as well:

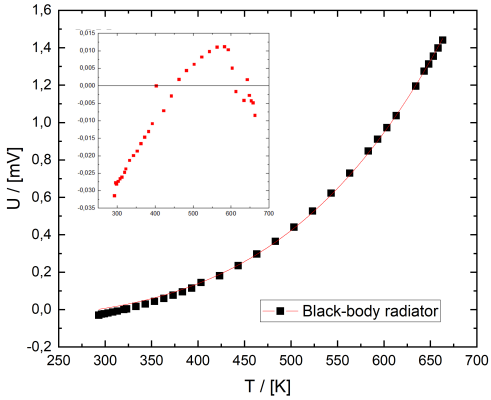

Figure 7: Measured thermoelectric voltage plotted against the temperature of the cavity radiator. A fit was adjusted to determine the exponent given in Tab. 4. Errors which are not plotted are too small for the chosen representation

Evaluation method	b
Fit in 7	4.077±0.052
Double-log in 8	4.073 ± 0.019

Table 4: Evaluated exponents b for the cavity radiator by two different methods.

The temperature and the thermo-electric voltage are plotted double logarithmically in Fig. 8 according to

$$ln(\frac{U}{1-(\frac{T_0}{T})^4}) = ln(c) + b\,ln(T) \qquad (8)$$

where c is a constant and b is the exponent to which the voltage U is proportional to. A fit was adjusted to the linear part for high temperatures. The values marked in grey in Fig. 8 were excluded for the evaluation of the power, since these values deviate significantly. This deviation occurs because of the usage of thermopile which is located in front of the cavity radiator. The end of the termophile measuring the reference temperature is not linked to a heat bath but measures the surrounding temperature, which is also influenced by the radiation of the cavity radiator. For small temperatures we may have a smaller absolute influence than for high temperatures but the relative influence is much higher as the temperature of the black body than for high temperatures. Subsequently the values for small temperatures are shifted down and we neglected those values in this evaluation method. The slope more specifically the exponent b is given in 4.
Due to the T_0 drift of the reference temperature already mentioned in the previous part of the experiment, the influence of the systematic error is also

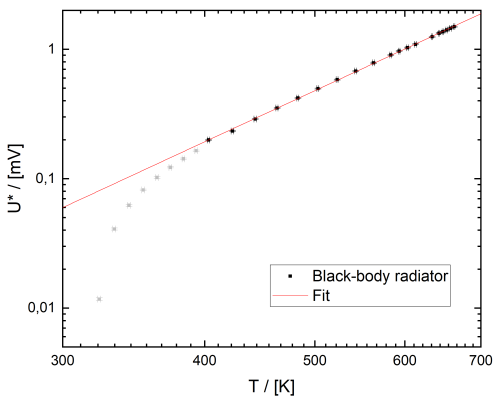

Figure 8: Double logarithmic plot of the voltage against temperature according to Eq. 3.2.1 to verify the Stefan-Boltzmann law. Errors not plotted are too small for the chosen representation.

given here. The drift increases with increasing temperatures in the cavity radiator. This could cause the deviation from the theoretical value in both methods. Furthermore, no new reference temperature was determined for this part of the experiment, but the one from 3.1.1 was used, which contributes to a falsification of the exponent b. Since the determined exponent only shows a deviation of 1.9% for the fit in 7 and 1.8% for the fit in 8 from $b = 4$, the Stefan-Boltzmann law is nevertheless considered to be fulfilled.

3.2.2 Comparison of the two evaluation methods

By comparing those two evaluation methods, the determined exponents b in 4 only show a deviation of 1.9% for the fit in Fig. 7 and 1.8% for the fit in Fig. 8 from $b = 4$. The deviation of the fit is a little bit higher than of the double-logarithmic evaluation. However, we cannot say that this method is better than the fit. Since in the double-logarithmic evaluation we assume that the measured temperature follows T^4 in advance in order to determine the exponent b. For the other method, it has to be said that its evaluation is numerical which optimises all not given parameters. The peril is that this evaluation ascertains the parameters wrongly as it just optimises the seemingly best fit. Consequently k and b in Eq.5 can differ when this method optimises in the wrong places. Nevertheless, considering that for the double-logarithmic method we have to assume that our measured voltage follows the Stefan- Boltzmann law in order to determine the exponent, we think that the numerical evaluated method is more accurate as the deviation resembles the double-logarithmic method strongly and does not need any assumptions the result in order to obtain it.

3.2.3 Lambert's law

Lambert's law states: If a radiation source or reflected radiation has a constant and direction-independent radiance on a surface, then the radiated power intensity decreases with decreasing angle φ.
To investigate this law in more detail, the radiated power of the cavity radiator was first measured with increasing angle from $\varphi = 0°$ to $\varphi = 50°$. In Fig. 9 the voltage measured by the thermopile as well as the theoretical curve are depicted against the angle φ. Fig. 9 shows a deviation of the voltages of large angles from the theoretical curve. The decrease of the thermoelectric voltage from $\varphi = 0°$ outgoing angles regarding to increasing angles results in the opening direction of the cavity radiator, as it points in the direction of $\varphi = 0°$. The cavity radiator also has a radiating direction, which is why the thermoelectric voltages suffer a downward drift at large angles. The height of the opening of the cavity radiator is below that of the thermopile, which is why this further distorts the measurement.
The drop in the relative voltages at large angles in Fig. 9 is further explained by the fact that at small angles the background radiation is more influenced by the cavity radiator than at large angles in regard to the radiation direction based on the opening direction. Due to the normalisation U/U_0 of the voltage, the falsification of the reference temperature by the cavity radiation attaches more and more weight to increasing angles. The measurement of the reference temperature is angle-dependent as well and, according to Lambert's law, less influenced at large angles than at small ones.

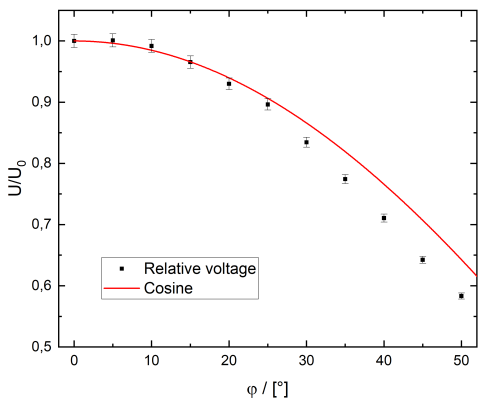

Figure 9: The relative voltage U/U_0 and the cosine is depicted in reference to the angular φ in order to compare the data with the theory of Lambert's law.

4 Summary

Using a cavity radiator and the Leslie cube, the Stefan Boltzmann law was proven in two different ways. Once with the Leslie cube and again with a cavity radiator.
With the Leslie cube, the emission properties of the four different sides were also examined. It was noticeable that the black and white sides behaved similarly in the infrared range, which suggests that these sides are made of the same material and that the visual properties of the colours have no influence. the visual properties of the colours have no influence whatsoever. On the basis of the nickel sides, however, it can be observed that the surface quality does have an influence on the radiation capacity. The matt and reflective sides, however, do not follow the Stefan Boltzmann law.
Subsequently, the behaviour of various filters exposed to infrared radiation was observed. The infrared transmittance τ of window glass, NaCl, silicium was calculated to be (2.0755±0.0012)%, (78.538 ± 0.038)%, (35.991±0.071)% respectively.
Finally, the validity of Lambert's law was checked. This was done with the cavity radiator used before. The measured values for the theoretical verification of the law are realistic. The orders in the graphs cover the assumed considerations and also the theory, which is why Lambert's law is considered confirmed.

References

[1] Geräteanleitung des Thermostats LAUDA Ecoline Staredition Re104

[2] Geräteanleitung Multimeter Agilent 34405A 5 1/2 Digit der Firma Agilent Technologies

[3] Physikalisches Grundpraktikum, Universität Würzburg, Modul C1, Versuch 33, Wärmestrahlung, 2021

[4] https://de.universaldenker.org/formeln/1150

Aufgerufen am: 01.10.2021 um 9.30 Uhr

[5] https://refubium.fu-berlin.de/bitstream/handle/fub188/6080/5kap6.pdf?sequence=6isAllowedy

Aufgerufen am: 01.10.2021 um 10.17 Uhr

[6] https://de.wikipedia.org/wiki/SiliciumPhysikalische Eigenschaften

Aufgerufen am: 01.10.2021 um 10.30 Uhr

[7] https://de.wikipedia.org/wiki/Natriumchlorid Physikalisch/ChemischeAnwendungen

Aufgerufen am: 01.10.2021 um 10.43 Uhr

5 Appendix

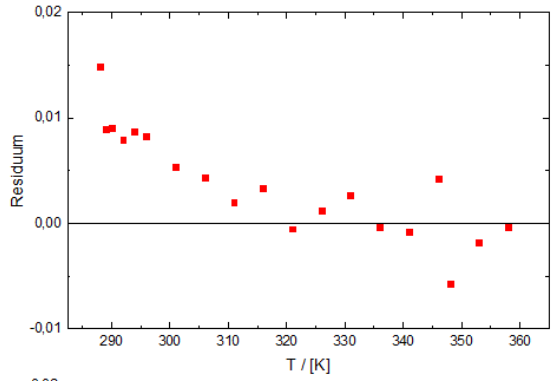

Figure 10: Residuum of the reflective side of the Leslie cube

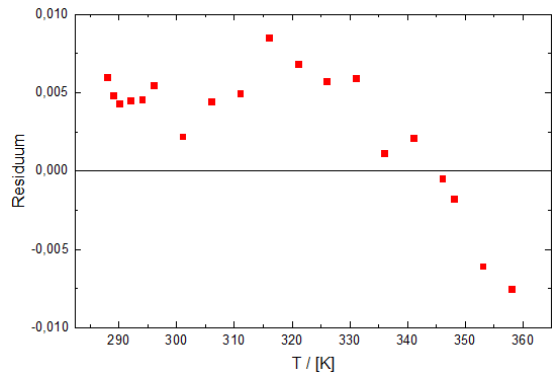

Figure 11: Residdum of the white side of the Leslie cube

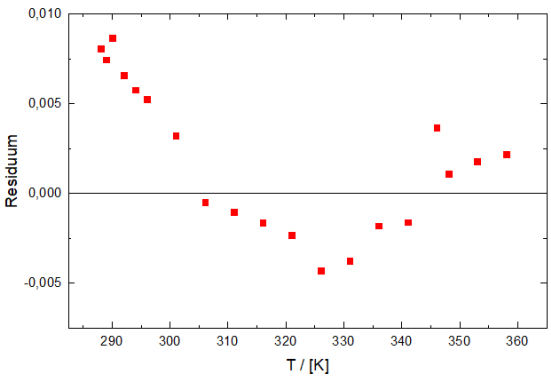

Figure 12: Residuum of the dull side of the Leslie cube